

Exercise (From HOTS): Four cards are shuffled and placed face down in front of you. Their faces (hidden) display the four elements: water, earth, wind and fire. You are to turn the cards over, one at a time, until you either win or lose. You win if you turn over water and earth. You lose if you turn over fire. What is the probability that you win?

Recognize that "wind" is irrelevant.

There are $3!$ possible orderings, they are ELO.

Two of them

(water, e, f) , (e, water, f)

have water and earth before fire.

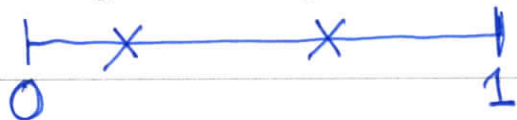
So, prob. of winning is $\frac{2}{6}$.

win

Or: ~~lose~~ if fire is in 3rd position

Since cards are equally likely to be in the three positions, $P(\text{win}) = \frac{1}{3}$

Exercise (From HOTS): Take a stick and break it randomly into three pieces (i.e., two randomly placed breaks on the stick). What is the probability you can form a triangle from the pieces?



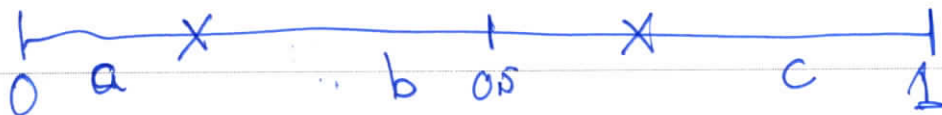
Assume positions are uniform between 0 and 1.

Can form a Δ iff sum of lengths of two smaller pieces is greater than the length of the largest



i.e. Cannot form a Δ iff the longest side has length of at least 0.5

i.e. Cannot form a Δ iff any of the three pieces has length at least 0.5

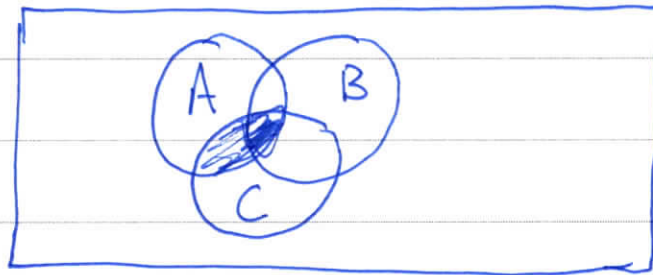


$A = \{ \text{segment } a \text{ is at least length } 0.5 \}$

$B = \{ \text{" } b \text{ " " " " " " } \}$

$C = \{ \text{" } c \text{ " " " " " " } \}$

$$\begin{aligned}
 P(\text{cannot form a } \Delta) &= P(A \cup B \cup C) \\
 &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\
 &\quad - P(B \cap C) + P(A \cap B \cap C)
 \end{aligned}$$

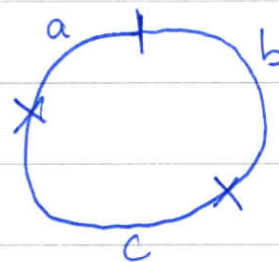


$$= P(A) + P(B) + P(C) \quad \text{since intersections have prob. 0}$$

$$\begin{aligned}
 \text{Now, } P(A) &= P(\text{segment } a \text{ is at least length } 0.5) \\
 &= P(\text{both breaks are greater than } 0.5) \\
 &= \frac{1}{4} \quad (\text{they are like coin flips})
 \end{aligned}$$

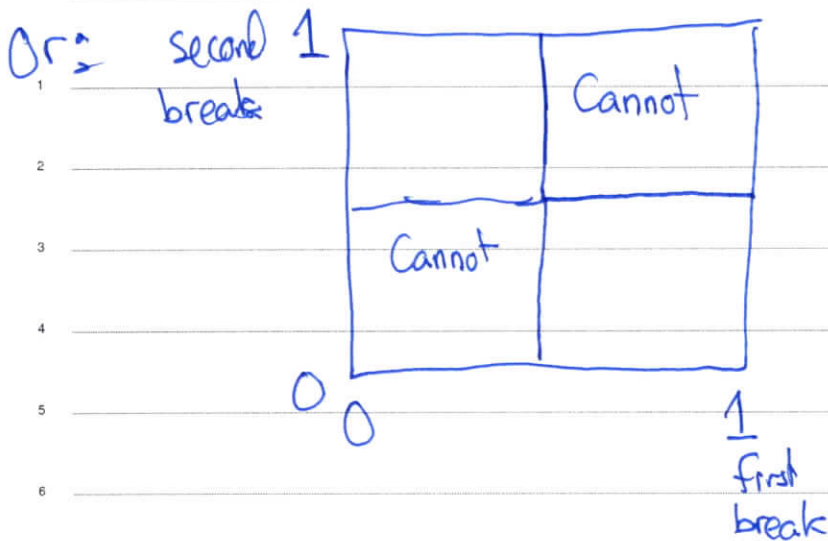
$$P(C) = \frac{1}{4}$$

$$P(B) = \frac{1}{4}$$



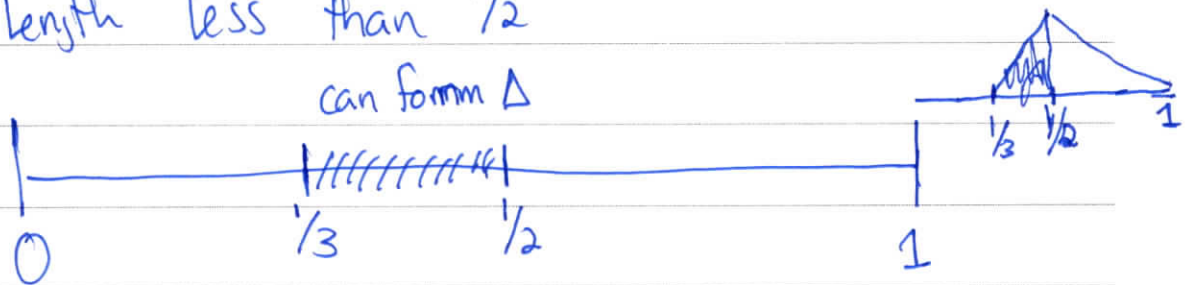
$$P(\text{cannot form a } \Delta) = \frac{3}{4}$$

$$P(\text{can form a } \Delta) = \frac{1}{4}$$



A solution?

The longest side has length between $\frac{1}{3}$ and 1.
Can form a Δ iff longest side has length less than $\frac{1}{2}$



$$P(\text{Can form a } \Delta) = \frac{\frac{1}{2} - \frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{4}$$

But: This is not correct because length of longest piece is not uniform over $(\frac{1}{3}, 1)$

Conditional Probability Rules

1. **Rule for Conditional Probability:** For any events A, B with $P(B) > 0$,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

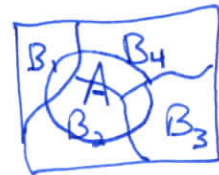
"A given B"

2. **Multiplication Rule:** For any events A, B ,

$$P(A \text{ and } B) = P(A \cap B) = P(A|B)P(B).$$

3. **Law for Total Probability:** Suppose events $B_1, B_2, \dots, B_n$ form a **partition** of Ω . Then for any event A ,

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i).$$



4. **Bayes' Theorem:** Suppose events $B_1, B_2, \dots, B_n$ form a partition of Ω .

Then for any event A ,

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}.$$

Exercise (Partially from HOTS): You are dealt exactly two playing cards from a well-shuffled standard 52-card deck. The standard deck contains exactly four Kings. (a) What is the probability that both of your cards are Kings? (b) What is the probability that the second card you are dealt is a King?

$$(a) \quad A = \{ \text{first card is king} \}$$
$$B = \{ \text{second card is king} \}$$

$$\begin{aligned} P(\text{both kings}) &= P(A \cap B) \\ &= P(A)P(B|A) \quad \text{using multiplication rule} \\ &= \left(\frac{4}{52}\right)\left(\frac{3}{51}\right) \end{aligned}$$

$$(b) \quad P(B) = \frac{4}{52}$$

Or:

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

using LFTP

$$\begin{aligned} &\text{since } A, A^c \text{ is a partition of } \Omega \\ &= \left(\frac{3}{51}\right)\left(\frac{4}{52}\right) + \left(\frac{4}{51}\right)\left(\frac{48}{52}\right) \\ &= \frac{4}{52} \end{aligned}$$

Exercise (From HOTS): You have a large jar containing 999 fair pennies and one two-headed penny. Suppose you pick one coin out of the jar and flip it ten times and get all heads. What is the probability that the coin you chose is the two-headed one?

$B = \{\text{choose the two-headed coin}\}$

$A = \{\text{all 10 flips are heads}\}$

Want $P(B|A)$. Use Bayes' Rule.
(Note that $P(A|B)$ is "easy".)

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$

(B, B^c is my partition)

$$= \frac{(1)\left(\frac{1}{1000}\right)}{(1)\left(\frac{1}{1000}\right) + \left(\frac{1}{2}\right)^{10}\left(\frac{999}{1000}\right)}$$

$$= 0.51$$

Exercise (Similar to question in HOTS): Polygraph tests (lie detector tests) are often routinely administered to employees or prospective employees in sensitive positions. According to studies of polygraph reliability (Gastwirth, J., 1987. The statistical precision of medical screening procedures, *Statistical Science*, 3, 213-222) if a person is lying, the probability that this is detected by the polygraph is 0.88, whereas if the person is telling the truth, the polygraph is indicating that he/she is telling the truth with probability 0.86. Now suppose that on a particular question the vast majority of subjects have no reason to lie, so that 99% tell the truth.

Suppose a subject produces a positive reading on the polygraph (hence indicating that he/she is lying). What is the probability that the polygraph is incorrect and that he/she is telling the truth?

$A = \{ \text{positive result from polygraph} \}$

$B = \{ \text{person telling the truth} \}$

Want $P(B|A)$. $P(A|B) = 1 - 0.86 = 0.14$

Bayes' Rule:

$$\begin{aligned}
 P(B|A) &= \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)} \\
 &= \frac{(0.14)(0.99)}{(0.14)(0.99) + (0.88)(0.01)} = 0.94
 \end{aligned}$$

Exercise (Very similar to a question in HOTS): I flip two coins hidden from your view.

Version One: You ask, "Is at least one of the coins heads?" I respond, "Yes."

What is the probability they are both heads (from your perspective)?

Version Two: I say "At least one is heads." What is the probability they are both heads (from your perspective)?

Version Three: You happen to see one of the coins, and notice that it is heads.

What is the probability they are both heads (from your perspective)?

Version One:

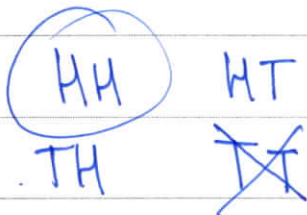
$A = \{\text{at least one heads}\}$

$B = \{\text{both heads}\}$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{using Rule for cond. prob.}$$

$$= \frac{P(B)}{P(A)}$$

$$= \frac{1/4}{1 - 1/4} = \frac{1}{3}$$



Version Two: Here the person not viewing the flips asks themselves: "How did the coin-flipper decide what info. to provide?"

A reasonable model for the info. provided is as follows

$N D$ <u>Result</u>	<u>Info.</u>	
HH	"at least one heads"	"notice heads"
HT	$\left\{ \begin{array}{l} \text{"at least one heads" w.p. } \frac{1}{2} \\ \text{"at least one tails" w.p. } \frac{1}{2} \end{array} \right.$	
TH		
TT		

Let $A = \{ \text{info. provided is "at least one heads"} \}$
 $B = \{ \text{both flips are heads} \}$

$$P(B|A) = \frac{P(B)}{P(A)}$$

$$\begin{aligned}
 P(A) &= P(A|HH)P(HH) + P(A|HT)P(HT) + P(A|TH)P(TH) + P(A|TT)P(TT) \quad \text{LFTP} \\
 &= (1)\left(\frac{1}{4}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) + (0)\left(\frac{1}{4}\right) \\
 &= \frac{1}{2}, \text{ so } P(B|A) = \frac{P(B)}{P(A)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}
 \end{aligned}$$